



Virtual SOC Capstone

Team Working Agreement

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1. Team Information

Members: Evan Ellsworth (Cybersecurity), Parker Sherman (Computer Science), Jay Hedding (Cybersecurity), Jack Klein (Computer Science)

Strengths & Skills

- Evan: Cybersecurity, SIEM visualizations, Linux, System Administration, Offensive and Defensive Technologies
- Parker: Project management, system setup, and architecture design
- Jay: Cybersecurity, Technical setup, VM familiarity, Linux, python
- Jack: Linux System knowledge and CLI Navigation, basic offensive security knowledge, basic networks knowledge, (python, javascript, c++, java, html, CSS/bootstrap)

Roles & Responsibilities

- Evan: Openstack management, client communication, network design, SIEM integration
- Parker: Project manager/upkeep of the git repo and issue creation
- Jay: Documentation and research, network configuring, tester
- Jack: Development and maintenance, tool configuration and setup.

2. Purpose & Goals

Mission Statement: Our goal is to build an isolated virtual SOC on Miami's OpenStack for teaching and demos. The comprehensive network will show real-time simulated attacks and defense using tools like firewalls, NIDS, EDR, and a SIEM to give students the opportunity to get hands-on experience.

Success Criteria:

- **Capability to simulate an attack:**
 - An external attacker (Kali Linux) should be able to simulate an attack against the internal systems
- **Network Isolation:**
 - The network of all the defined systems should be completely isolated from the Miami Network once everything needed is downloaded
- **Detection of attacks**
 - NIDS should be able to detect attacks made by an external attacker and report them to the SIEM
 - Logs of these attacks should also properly be stored
- **Automation of Deployment**
 - The entire environment should be able to be deployed via a script with all functionality included
 - A well-documented set of steps should be made to walkthrough the setup, usage, and teardown of the system
- **Reliability and Usability**
 - All systems in the environment should work reliability without the user having to troubleshoot any bugs
 - There should be minimal set up for anybody that wishes to use the system

3. Meetings & Values

Meeting Frequency

- Weekly Team Meeting
- Weekly Client Meeting
- Biweekly Instructor Meeting

Sprint Schedule:

- Sprint 1: 9/08-9/21
- Sprint 2: 9/22-10/05
- Sprint 3: 10/06-10/19
- Sprint 4: 10/20-11/02
- Sprint 5: 11/03-11/16
- Sprint 6: 11/17-12/07

Core values:

- Communication: As a team communication is most vital to producing efficient work
- Respect: As a team we will show each other equal respect
- Consistency: It is important that everyone remains present in order to allow for effective communication, and allow us to move forward at a constant pace each sprint.
- Accountability: We share ownership over the result of this project, so team members will be held accountable.

Everyone is expected to act according to our team values in how they operate as a member of the team. Missing a deadline is alright, but there should be real work always being done each sprint. Feedback should be brought up in a meeting if someone has something to sort out. Group or personal messaging is also acceptable.

4. Practices & Tools

Definition of Done:

- Well Documented: Code has documentation and networking has high quality lab notebooks to accompany it
- Functional and security tests completed
- Task management updated in gitlab

Tools & Repositories:

- Gitlab: we will update gitlab to track progress over time and to manage the sprint workflows
- Openstack: We will be hosting our product on a dedicated domain in OpenStack
- Tools: We will be using Suricata NIDS, Wazuh XDR/SIEM, pfSense firewall

5. Communication

Work outside of meetings will be almost entirely asynchronous. Meetings will take place either in person, or online over Webex/Discord. Team members are expected to respond to communication within 24 hours.

Communication Channels:

- Email (w/ Client)
- Webex
- Discord
- Mobile

6. Decision-making & Escalation

Decision Process:

- Consensus: Decisions will be approved if agreed on by the majority
- Dissent: Team members with objections should be able to suggest an alternative

Escalation Path:

- Based on the context of the issue, the action taken will be either to conclude with majority rule, or to seek advice from the client (Dr. Walden).

7. Quality & Deadlines

Issues that are placed in certain sprints are expected to be completed by the end of the sprint. Delivered items should be fully functional according to the goals of the task/issue, and should be well documented and readable. Configuration/setup style deliverables should come with documentation through complementary lab notebook entries. Notebook documentation should provide correct paths to mentioned objects. Deliverables will additionally be tested and read by other group members before considered done. How a missing deadline is handled will be context dependent, and handled through meeting or messaging by the team.

8. Conflict Resolution

First and foremost, as a part of our team values, team members should practice emotional maturity, neutrality, respect, and positivity especially in respect to issues or dissenting opinions. It is important to communicate effectively and maturely in person or through messaging. Communication should precede action, and it is not acceptable to keep a chip on your shoulder. All team members should agree to approach conflict with respect and neutrality, deescalate, and work towards a solution collaboratively.

9. Signatures

Jay Hedding 9/20/25

Evan Ellsworth 9/20/25

Parker Sherman 9/20/25

Jack Klein 9/20/25

Review date: 10/03/2025